COMP4035 Database System Implementation (2025-26)

Coding Project: B+ tree

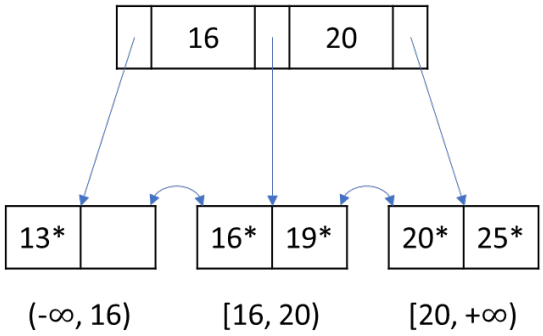
Project Requirement

Write a program to solve the B+ tree insertion & deletion exercise:

- Given an initial B+ tree and a sequence of insertion and deletion operations, run a program to produce the final tree.

Demo question:

Initial state, d = 1;
Q1. Insert 27*
Q2. Insert 14*
Q3. Insert 22* and 21*
Q4. Delete 21*
Q5. Delete 22*
Q6. Delete 20*
Q7. Delete 25*



Your B+ tree program's logic and output must follow the insertion and deletion rules taught in this course. See the "b+ tree exercise.pptx" slides on Moodle for a detailed example. In short, treat this like a written-exam question: you should be able to draw the final tree from your program's output. You will pass only if that drawn tree would be marked correct in the written exam.

General Guidelines

Objectives:

- This project is task-oriented. The key criterion is whether your program correctly solves the classic B+ tree exercise described above.
- The process is flexible. You may choose any reasonable input format and interaction method for your program.

Coding:

- Implement your program in Python, C, C++, or Java.
- This is an individual project; no teamwork.

LLM Tools:

- You may use LLM tools for assistance.
- You must understand the code and be able to modify it.
- You are responsible for ensuring your program's output is correct.

Presentation Guidelines

Time: Nov 27, 10:30am - 11:30am

During the final lecture on November 27, Prof. Yang will provide a course review in the first hour. The presentation session will begin at approximately 10:30 a.m.

Format: In-person or online.

- In-person presentation is strongly recommended.
- The online ZOOM presentation is a make-up option and is **not recommended**. It is more challenging and may receive a lower score than the in-person format.

Key points for online presentation:

- Any student may apply for an online presentation with a reasonable reason.
- Application procedure:
  1. Email Dr. Piao (czpiao@comp.hkbu.edu.hk) with your reason on or before Nov 23 (Sun), 23:59.
  2. Receive a confirmation email from Dr. Piao.
  3. Your name will be added to the online presentation list on Moodle.
- The Zoom link will be posted on Moodle on the morning of Nov 27. Detailed arrangements will be released later based on the number of online presentations.

Evaluation:

- During your presentation, you must run your program to solve one of the three tasks below. Your project will be graded on whether your program solves the chosen task correctly.
- For tasks 2 and 3, you need to modify your code during the presentation session.

Task	Level	Description	Grade
1	Basic	The demo question.	C
2	Normal	A similar question to the demo with a larger d (e.g., d = 2).	B
3	Challenging	A question with a different B+ tree setting (e.g., LRU -> MRU in Lab 2).	A

Output Quality	Output Format	Grade
Good	The program outputs the tree in text. You then manually draw the tree from the output.	C/B/A (task level)
Excellent	The program visualizes the tree structure after each step in a reasonable format.	C+/B+/A+ (add a '+')

Tentative score mapping for the coding project:

Grade Letter	A	B	C	+	-
Score	100	70	35	+10	-10

Presentation Procedure:

1. Tasks 2 and 3 will be released at the start of the presentation session. Choose one to present.
2. When you are ready, raise your hand. The instructor or tutor will come to check. Note: the tutor will only check the demo question.
3. During presentation:
    a. Use your personal laptop or the lab PC.
    b. Manually convert the initial tree and update operations into valid input data (you decide the input format).
    c. Run your program to produce the final tree.
    d. If your program cannot visualize the tree, manually draw the tree based on the program's output.
    e. Evaluation focuses on program output, but if the instructor suspects excessive reliance on LLMs, **you may be asked to explain your code**. Your grade may be downgraded (e.g., B → B−).
    f. You will receive a grade letter immediately after your presentation. If your first attempt fails, you may try again if time permits. The first attempt has priority, and a second attempt may be refused due to the 11:30 time limit.

Submission:

- After the presentation, submit your code and a brief running guideline on Moodle (how to run your program, input/output format for the B+ tree exercise).
- These are not graded and are for backup only, but failing to submit will incur a penalty (e.g., B → B−).
- Due: Nov 30, 23:59.